

Formula de convoluție

Fie X_1 și X_2 două variabile aleatoare continue independente și $Y = X_1 + X_2$.

Atunci:

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X_1}(y-x_2) \cdot f_{X_2}(x_2) dx_2 = \int_{-\infty}^{\infty} f_{X_2}(y-x_1) \cdot f_{X_1}(x_1) dx_1$$

Exemple

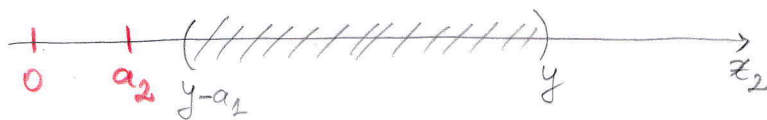
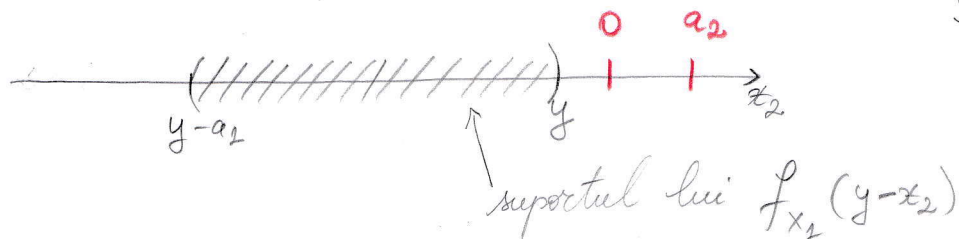
[1] $X_1 \sim \text{Unif}(0, a_1)$, $X_2 \sim \text{Unif}(0, a_2)$, X_1, X_2 independente, iar $0 < a_1 < a_2$ și $Y = X_1 + X_2$

$$f_{X_i}(x) = \begin{cases} \frac{1}{a_i}, & x \in (0, a_i) \\ 0, & x \notin (0, a_i) \end{cases} \quad i = \overline{1, 2}$$

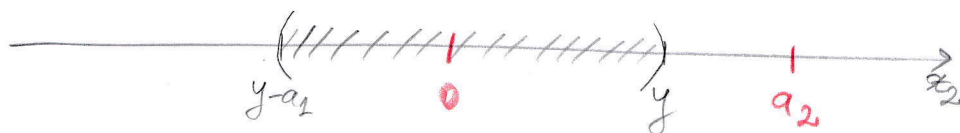
$$f_{X_1}(y-x_2) = \begin{cases} \frac{1}{a_1}, & y-x_2 \in (0, a_1) \\ 0, & \text{în rest} \end{cases}, \quad f_{X_2}(y-x_2) = \begin{cases} \frac{1}{a_2}, & x_2 \in (y-a_1, y) \\ 0, & \text{în rest} \end{cases}$$

$$y-x_2 \in (0, a_1) \Leftrightarrow 0 < y-x_2 < a_1 \Leftrightarrow -a_1 < x_2-y < 0 \Leftrightarrow y-a_1 < x_2 < y \\ \Rightarrow x_2 \in (y-a_1, y)$$

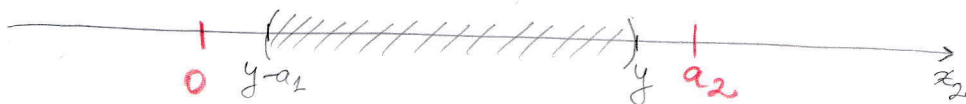
$$\text{Atunci } f_Y(y) = \int_{-\infty}^{\infty} f_{X_1}(y-x_2) \cdot f_{X_2}(x_2) dx_2 = \frac{1}{a_2} \cdot \underbrace{\int_0^{a_2} f_{X_1}(y-x_2) dx_2}_I$$



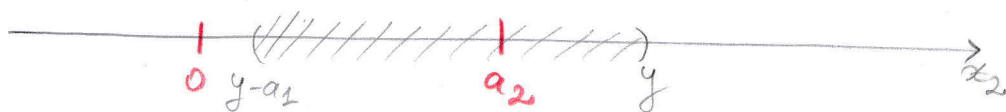
I Dacă $y < 0$ sau $y \geq a_1 + a_2$ atunci $I = 0$



II Dacă $y \in [0, a_1]$ atunci $I = \int_0^y \frac{1}{a_1} dx_2 = \frac{1}{a_1} \cdot y$



III Dacă $y \in (a_1, a_2]$ atunci $I = \int_{y-a_1}^y \frac{1}{a_1} dx_2 = \frac{1}{a_1} \cdot (y - (y - a_1)) = 1$



IV Dacă $y \in (a_2, a_1 + a_2)$ atunci $I = \int_{y-a_1}^{a_2} \frac{1}{a_1} dx_2 = \frac{1}{a_1} \cdot (a_2 - (y - a_1))$

Asadar $f_y(y) = \begin{cases} 0, & y < 0 \text{ sau } y \geq a_1 + a_2 \\ \frac{y}{a_1 \cdot a_2}, & y \in [0, a_1] \\ \frac{1}{a_2}, & y \in (a_1, a_2] \\ \frac{a_1 + a_2 - y}{a_1 \cdot a_2}, & y \in (a_2, a_1 + a_2) \end{cases}$

[2] Die $X_1, X_2 \sim \text{Norm}(0, 1)$ unabhängige, $Y = X_1 + X_2$.

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X_1}(y-x_2) \cdot f_{X_2}(x_2) dx_2$$

$$f_{X_i}(x_i) = \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{x_i^2}{2}}, \quad x_i \in \mathbb{R}, \quad i = \overline{1, 2}$$

$$f_Y(y) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{(y-x_2)^2}{2}} \cdot \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{x_2^2}{2}} dx_2$$

$$f_Y(y) = \frac{1}{2\pi} \cdot \int_{-\infty}^{\infty} e^{-\frac{y^2 - 2x_2y + 2x_2^2}{2}} dx_2 = \frac{1}{2\pi} \cdot \int_{-\infty}^{\infty} e^{-\frac{2(x_2 - \frac{1}{2}y)^2}{2}} dx_2$$

$$f_Y(y) = \frac{1}{2\pi} \cdot \int_{-\infty}^{\infty} e^{-\left(x_2 - \frac{1}{2}y\right)^2} \cdot e^{-\frac{1}{4}y^2} dx_2 = \frac{e^{-\frac{1}{4}y^2}}{2\pi} \cdot \int_{-\infty}^{\infty} e^{-\left(x_2 - \frac{1}{2}y\right)^2} dx_2$$

s.v. $x_2 - \frac{1}{2}y = t \Rightarrow dx_2 = dt$

$$x_2 \rightarrow -\infty \Rightarrow t \rightarrow -\infty$$

$$x_2 \rightarrow \infty \Rightarrow t \rightarrow \infty$$

$$f_Y(y) = \frac{e^{-\frac{1}{4}y^2}}{2\pi} \cdot \int_{-\infty}^{\infty} e^{-t^2} dt = \frac{e^{-\frac{1}{4}y^2}}{2\pi} \cdot \sqrt{\pi} = \frac{e^{-\frac{1}{2} \cdot \left(\frac{y-0}{\sqrt{2}}\right)^2}}{\sqrt{2\pi} \cdot \sqrt{2}}$$

$\Rightarrow \underline{Y \sim N(0, 2)}$

$$\int_{-\infty}^{\infty} e^{-t^2} dt = \sqrt{\pi}$$

Integrala
Euler - Poisson

μ
 σ